Chapter 10
Research Integrity

In Chaps. 5, 6, 7, 8, and 9 I have focused on specific issues related to research with human subjects, such as informed consent, confidentiality, risks, benefits, and vulnerability. In this chapter, I will shift gears and examine a topic that indirectly impacts human subjects but which is nevertheless very important: research integrity. Research integrity (or responsible conduct of research, RCR) has to do with following ethical and legal standards in the conduct of research (Shamoo and Resnik 2015). These include rules pertaining to research with human and animals subjects as well as those concerning the conduct of science itself, such as norms for recording, reporting, analyzing, sharing, publishing and interpreting data; assigning authorship; disclosing and handling conflicts of interest; working with collaborators, students, and trainees; reviewing manuscripts and grants; managing financial and other resources; and investigating allegations of misconduct (Shamoo and Resnik 2015). This chapter will consider some RCR issues which have an important bearing on research with human subjects. But first, I will explain why investigator integrity is essential to research with human subjects.

10.1 The Importance of Integrity in Research with Human Subjects

Investigator behavior is probably the single most important factor in protecting the rights and welfare of human subjects. Regulations and guidelines have little utility if investigators ignore them. IRB review and oversight is also ineffective if the investigator does not comply with the approved protocol and consent process as well as ethical and legal standards for the conduct of research. Investigators (and research staff members) have direct contact with human subjects and therefore have the greatest opportunity to either help or harm them. IRBs, institutions, and sponsors must therefore trust that investigators will act responsibly. Chapter 2 described

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several cases (e.g. Poisson, Poehlman, Hwang, and Wakefield) in which investigator misbehavior harmed human subjects or placed them at undue risk.

A recent episode at Duke University demonstrates how misconduct can directly harm research subjects. In November 2015, the Office of Research Integrity (ORI), which oversees the integrity of NIH-funded research, found that former Duke University investigator Anil Potti had included false data in one NIH grant application and 9 published papers. As part of the settlement, ORI required Potti to retract the papers and agree to supervision of his federally funded research for a period of 5 years (Office of Research Integrity 2015). Earlier that year, Duke University had settled a lawsuit brought by eight participants in Potti's oncology clinical trial who claimed they were harmed by his fraudulent research (Stancil 2015).

Potti had been working with Joseph Nevins on cancer genomics in the early 2000s. They hypothesized that one could use a tumor's genomic signature to predict its response to chemotherapy. In 2006, they published a paper in *Nature Medicine* describing a predictive model that they had developed using computer programs and publicly available genomic data for the tumors (Potti et al. 2006). Shortly after the paper appeared in print, two statisticians, Keith Baggerly and Kevin Coombes, detected errors in the model and could not replicate the results. They contacted Potti and Nevins about issues they had with the paper and published a critique in the same journal (Coombes et al. 2007). Potti and Nevins responded to these critiques and insisted that their model worked. In 2008, Bradford Perez, a 3rd year medical student at Duke University who was working under Potti and Nevins, found problems with their predictive model and discovered that it had not been independently validated. Perez was worried that patients could be harmed because Potti and Nevins were preparing to begin clinical trials based on their research. Perez shared his concerns with university administrators, who encouraged him to work with Potti and Nevins to deal with these issues. They expressed the view that the problems he found amounted to a scientific disagreement rather than misconduct and warned him that he could damage his own career if he brought charges against his supervisors (Goldberg 2015). In 2009, Baggerly and Combes learned that clinical trials had begun at Duke. They published another critique of Potti's research and expressed their concerns to the Duke IRB. The IRB temporarily suspended the clinical trials in order to review the research but then allowed them to start again. In July 2010, *The Cancer Letter* reported that Potti had falsely stated he was a Rhodes Scholar on his curriculum vita and grant applications submitted to the NIH and American Cancer Society (Goldberg 2015). Shortly thereafter, the IRB suspended Potti's clinical trials again and Duke launched a misconduct investigation. In 2012, Potti resigned his position at Duke and took a job at the North Dakota Cancer Center.

Looking beyond this extraordinary case, harms to human subjects or rights violations can also occur as a result of less dramatic transgressions, such as:

- Enrolling a subject in a study who does not meet the inclusion criteria;
- Using unqualified staff to perform research procedures;
- Performing an inadequate review of the literature related to one's research;
- Using an invalid methodology or experimental design;

- Failing to adequately protect confidentiality;
- Not disclosing important information during the consent process;
- Failing to carefully monitor the health of subjects in a clinical study;
- Not reporting adverse events or other problems in a timely fashion.

Investigator behavior also plays a key role in promoting trust in research with human subjects. The cases discussed in Chap. 2 show how misbehavior that harms human subjects can undermine the trust that human subjects, communities, and the members of public place in investigators, institutions, sponsors, and the scientific enterprise. Societies have developed regulations, guidelines, and oversight systems to control investigator behavior and thereby protect human subjects.

Misbehavior that does not directly harm human subjects may nevertheless negatively impact society's trust in research if it leads to fraudulent, erroneous, or biased data or results that cause harm to society or increase the risk of harm. For example, Wakefield's study linking the MMR vaccine to autism lowered vaccination rates in the U.K. and other countries, and therefore may have increased infections of measles or other childhood diseases for which there is a vaccine. Merck's exclusion of some cardiovascular risk data from the VIGOR study may have harmed people who took Vioxx because they did not know some of the risks of the drug. If these risks had been fully disclosed to the public, fewer people may have taken the drug and, therefore, fewer may have been harmed. (See fuller discussion of this case below.) Although Poisson's alteration of his patient's medical records did not skew the overall results of the clinical trial, it easily could have if he had changed more records. Conflicts of interest (discussed below) are a significant concern in research because they may increase the risk of bias or misbehavior (Shamoo and Resnik 2015).

Misbehavior that does not harm human subjects or society may still undermine trust if it produces useless results or wastes public resources. One of the main justifications for exposing human subjects to risks is that the study is expected to yield knowledge that benefits society (Shamoo and Resnik 2015). If investigators fabricate or falsify data, then their research may not produce useful results, which would undercut its ethical justification and waste public resources. Mismanagement of research funds can also waste public resources and undermine trust.

Misbehavior may also erode investigators' trust in each other, which is crucial for the advancement of science (Whitbeck 1995). Research involves many types of social interactions between and among scientists (Hull 1988). Some of these include: working together in laboratories or research groups, collaborations among researchers in different groups, peer review, and publication. For these social interactions to work, scientists must be able to trust each other to follow certain standards of behavior (Shamoo and Resnik 2015). They must be able to trust that: they can rely on published research results; collaborators will not steal their ideas, data, or methods; they will receive proper credit for their accomplishments; reviewers will evaluate their work fairly and respect the confidentiality of the peer review, etc.

10.2 Misconduct in Research

The most serious violations of ethical or legal standards in research are classified as misconduct (Shamoo and Resnik 2015). The U.S. government defines research misconduct as: "fabrication, falsification, or plagiarism in proposing, performing, or reviewing research, or in reporting research results…misconduct does not include honest error or differences of opinion (Office of Science and Technology Policy 2000:76262)." This definition, which is accepted by numerous federal agencies, including the NIH, NSF, and FDA, focuses on three types of misbehavior: fabrication, falsification, or plagiarism (FFP). Fabrication is "making up data or results and recording or reporting them," falsification is "manipulating research materials, equipment, or processes, or changing or omitting data or results such that the research is not accurately represented in the research record," and plagiarism is "the appropriation of another person's ideas, processes, results, or words without giving appropriate credit (Office of Science and Technology Policy 2000:76262)."

Federal agencies decided to focus on FFP because they viewed broader definitions of misconduct, some of which included the category "other serious deviations" as too vague (Resnik 2003a). However, other countries use definitions of misconduct that go beyond FFP. Resnik et al. (2015c) studied the science policies of the top 40 countries ranked by total research funding. 22 (55%) of these countries had national misconduct policies. While all of these countries included FFP in the definition of research misconduct, most went beyond these categories and included other types of misbehaviors, such as: unethical authorship other than plagiarism (54.6%); unethical publication practices, such as duplicate publication (36.4%); failing to disclose a significant conflict of interest (36.4%); unethical peer review (31.8%); interfering with a misconduct investigation (27.3%), poor record-keeping (27.3%); forms of deception other than FFP, such as manipulating data (27.3%); serious deviations (22.7%); violating confidentiality (22.7%); and serious violations of human or animal research regulations (22.7%). In a separate study of 183 of the top U.S. academic institutions ranked by total research funding, Resnik et al. (2015b) found that 59% use a definition of research misconduct that goes beyond FFP. Included in these definitions were: other serious deviations (45.4%); significant violations of research regulations, such as animal or human subjects regulations (23.0%); misusing confidential information in peer review or other aspect of research (15.8%); interfering with a misconduct investigation (14.8%); unethical authorship other than plagiarism (14.2%); other forms of deception, such as data manipulation (13.1%); and theft or misappropriation of property (10.4%).

Misconduct investigations are legal proceedings in the U.S. and other countries. The ORI has developed policies and procedures that institutions are required to follow when they perform misconduct investigations involving NIH-funded research, but most institutions apply these to all research, regardless of the funding source (Shamoo and Resnik 2015). Research institutions are required to have a research integrity official (RIO) who receives formal allegations of misconduct. If the RIO determines that an allegation meets the definition of misconduct and is genuine, he

or she will appoint a committee to conduct an inquiry to determine whether there is enough evidence to support an investigation. If the inquiry committee concludes that an investigation is warranted, the RIO will appoint another committee to conduct an investigation. The investigation committee will obtain evidence by reviewing research records and other materials and interviewing witnesses. The committee will decide whether the evidence shows that the accused party (i.e. the respondent) has committed misconduct. A finding of misconduct must be supported by a preponderance of evidence, i.e. probability > 50%. Misconduct must be committed intentionally, knowingly, or recklessly (Office of Science and Technology Policy 2000). The RIO will act on the Committee's findings and recommend a punishment if misconduct is found. Sanctions may include: termination of the respondent's employment, demotion, increased supervision of the respondent's work during a probationary period, education in RCR, or a letter of reprimand.

If the research is federally-funded, the institution will report its findings to the appropriate agency. The ORI reviews findings for NIH-funded research. The federal agency may accept the institution's findings, ask for additional evidence or clarification, or conduct its own investigation. If the agency reaches a finding of misconduct, it will usually require the investigator to forego federal research funding for a period of time (sometimes indefinitely) and to correct the publication record (if appropriate) (Shamoo and Resnik 2015). Although researchers rarely face criminal or civil penalties for misconduct, Eric Poehlman served a year and a day in a federal prison and was required to pay the government $180,000 (see Chap. 2).

Institutions are required to follow rules for legal due process when conducting misconduct inquiries and investigations. Respondents are permitted to obtain legal counsel, interview witnesses, and review the evidence. They can also appeal misconduct findings to a federal court. To protect the reputations of the parties, institutions are required to maintain confidentiality during misconduct proceedings. However, sometimes deliberations or findings are unlawfully leaked to the press. ORI publishes cases on its website once it has made official findings of misconduct (Shamoo and Resnik 2015).

Scientific journals have adopted policies for dealing with misconduct allegations (Resnik et al. 2009, 2015d). However, their legal authority to investigate allegations of misconduct is limited. Editors will usually refer such matters to the authors' institutions for resolution. While an investigation is ongoing, the editors may publish an expression of concern, pending the outcome. If there is an official finding of misconduct, editors will usually retract the article, with or without the permission of the authors. Editors may face difficult ethical dilemmas when dealing with alleged misconduct. On the one hand, they have a duty to the journal's reputation and the integrity of the research record. On the other hand, they have a duty to not falsify accuse innocent people of wrongdoing. They may also be concerned about being sued for libel by authors. The Committee on Publication Ethics (2012) has developed some guidelines for editors and journals for handling misconduct cases.

The incidence of misconduct in research is thought to be rare. Fanelli (2009) conducted a systematic review of 21 misconduct surveys and found that 1.97% of scientists had admitted to fabricating or falsifying data or committing plagiarism at

least once in their careers. However, 14.2% said they had observed someone else committing misconduct. Both of these percentages may not reflect the true rate of misconduct. The self-reported rate (1.97%) may underestimate the actual rate because researchers may not be willing to admit that they have committed misconduct, even if their confidentiality is protected. The observed rate (14.2%) may overestimate the actual rate of misconduct because people who claim to have observed misconduct may not have enough evidence to determine whether it has occurred, and some respondents to surveys may observe the same act of misconduct, which may result in double-counting. Even if the rate of misconduct is as low as 2% or less, it still represents a serious threat to the integrity and trustworthiness of research that needs to be addressed (Shamoo and Resnik 2015).

10.3 Noncompliance

The Common Rule requires institutions to have written procedures for prompt reporting of "unanticipated problems involving risks to subjects or others or any serious or continuing noncompliance (45 CFR 46.103b5)." Serious or continuing noncompliance could qualify as research misconduct under federal policies if it involves FFP. As noted above, many academic institutions and some countries classify serious noncompliance with human research regulations as misconduct (Resnik 2015a, b). The NIH's intramural human research protection program defines serious noncompliance as noncompliance that increases the risks of the study or causes harm, decreases benefits, compromises the integrity of the research, or invalidates study data (National Institutes of Health 2016b). Some types of serious noncompliance could include: conducting human subjects research without obtaining IRB approval or when approval has expired, failing to obtain informed consent, significantly deviating from an IRB-approved protocol, fabricating or falsifying data, poorly supervising staff members, or breaching confidentiality.

Investigations of noncompliance in research with human subjects are administrative proceedings which are not nearly as legalistic as misconduct investigations. Anyone involved in research (i.e. investigators, staff, subjects) may report noncompliance. Investigators often report their own noncompliance, which they may discover through a self-audit or a report from a staff member. External audits (routine or for-cause) may also uncover noncompliance (National Institutes of Health 2016b). The IRB receives reports of noncompliance and investigates allegations. The IRB may obtain assistance from an ad hoc committee, administrative staff, or special experts (National Institutes of Health 2016b). If the noncompliance involves an allegation of research misconduct, then it will also be referred to the RIO for review, as occurred in the Potti case (above). The IRB is authorized to take the steps necessary to protect human subjects from potential harm while it is reviewing noncompliance issues. For example, the Duke IRB temporarily suspended Potti's clinical trial to protect human subjects while it was reviewing the research. If the IRB makes a find-

ing of noncompliance, it may recommend further action, such as: terminating or suspending the study; amending the study; additional education or training for the investigator or staff; requiring the investigator to develop policies and procedures to prevent noncompliance from occurring again; or requiring the investigator to inform subjects about the noncompliance if it impacts them. An investigator may ask an IRB to reconsider its findings or disciplinary actions (National Institutes of Health 2016b).

10.4 Whistleblowing in Research

Although federal regulations and institutional policies mandate protection for individuals who report noncompliance, misconduct, or other serious problems with research, whistleblowers still may encounter difficult ethical and personal dilemmas when deciding whether to make allegations (Malek 2010). Even if a whistleblower is not fired or demoted, he or she may face intimidation or harassment from supervisors, colleagues, or administrators. He or she may also develop a reputation as a troublemaker. Institutional officials who do not intimidate or harass whistleblowers may still ignore or minimize their concerns, which occurred when Perez raised issues with Potti's research. If a whistleblower is a graduate student or postdoctoral fellow whose research is supported by grants, he or she may lose funding if his or her supervisor is found to have committed misconduct or serious noncompliance. In some cases, a whistleblower may need to transfer to another laboratory or academic institution to continue his or her research (Malek 2010). Even if whistleblowers are not likely to suffer any adverse career consequences from reporting misconduct or noncompliance, they still may be wary of making an allegation because they want to avoid the stress and time-demands associated with the investigation. Misconduct investigations can last for several years and can be stressful for all parties involved, not just the respondents (Shamoo and Resnik 2015).

Because institutions have a great deal at stake in misconduct cases, such as their reputations or research funding, there is an inherent conflict of interest (discussed below) when they investigate misconduct. University officials may be hesitant to bring charges of misconduct against well-established, prestigious investigators because they do not want to lose grant dollars or face public backlash (Shamoo and Resnik 2015). Although regulations and policies require that members of committees investigating misconduct should have no conflicts of interest, conflicts be difficult to avoid, especially when many of the people involved know each other personally and professionally. To encourage whistleblowers to report allegations of misconduct or noncompliance, many institutions permit anonymous reports. However, it is not be possible to guarantee anonymity if the whistleblower is asked to testify as a witness (Shamoo and Resnik 2015).

10.5 Preventing Misconduct and Noncompliance

While it is important to report and investigate misconduct and noncompliance, prevention is even more important. Prevention can head-off problems before they occur and avoid the damage to human subjects, investigators, institutions, and the scientific enterprise that often results from significant violations of ethical or legal standards in research. To understand how to prevent misconduct and noncompliance, it is important to be familiar with its causes. Some of the causes of misconduct and non-compliance include (Institute of Medicine 2002a, b; Shamoo and Resnik 2015; Dubois et al. 2016):

- Pressures to produce results or publish;
- Poor supervision of subordinates;
- Poor communication;
- Lack of awareness or understanding of research rules;
- Lack of education or mentoring in research ethics and compliance;
- Conflicts between different roles in research (e.g. clinical vs. scientific);
- Conflicts of interest (see discussion below);
- Cultural differences concerning ethical standards;
- Personal conflicts among researchers and staff;
- Poor leadership.

Some forms of prevention may therefore include:

- **Policy development**: Institutions should develop policies related to research integrity issues. The policies should define expected standards of behavior and establish procedures for reporting, investigating, and adjudicating misconduct, noncompliance, and other significant problems in research. Institutions should inform investigators and research staff about their policies and keep them abreast of changes. They should also publicize their policies on websites (Institute of Medicine 2002a, b; Shamoo and Resnik 2015).
- **Education**. Institutions should educate investigators, research staff, students, trainees, and administrators about RCR topics, such as misconduct, compliance, data management, conflicts of interest, etc. RCR education and training should take place when an individual takes a position at an institution or matriculates, as well as at regular intervals subsequently (e.g. annual updates). Researchers who work with human subjects should have special instruction in the relevant regulations, policies, and guidelines (Institute of Medicine 2002a, b; Shamoo and Resnik 2015).
- **Mentoring**. Investigators mentor their students and trainees in research ethics. Mentoring involves teaching by example and through informal discussions. For example, a clinical investigator could mentor a medical student by showing her how to keep research records or explaining to her how the IRB submission process works. Junior faculty members can also benefit from mentoring from senior faculty (Institute of Medicine 2002a, b; Shamoo and Resnik 2015).

- **Leadership**. Institutional leaders should affirm their commitment to research ethics and compliance through their words and deeds. They should stress the importance of following ethical and legal standards for the conduct of research in their communications with investigators, students, trainees, staff, and administrators. Leaders who set a bad example through their questionable behavior or indifference to research integrity may encourage corruption, misconduct, noncompliance, and other ethical and legal problems (Shamoo and Resnik 2015).
- **Auditing**. Institutions should regularly audit research records, including data, case reports, and informed consent documents. Auditing can discourage misconduct and noncompliance and catch problems at the early stages, before they become worse (Shamoo 2013; Shamoo and Resnik 2015).
- **Administrative support**. Institutions should provide adequate administrative support for research ethics and compliance activities, including staff members who work with the IRB as well as officials who can consult with researchers, students, and trainees on ethical or legal issues. Many institutions have an upper-level administrator whose job is to oversee ethics and compliance activities (Institute of Medicine 2002a, b; Shamoo and Resnik 2015).
- **Accreditation**. The Association for Accreditation of Human Research Protections Programs (AAHRPP) accredits institutions which conduct, support, or review research with human subjects. AAHRPP has established standards pertaining to research oversight, policy development, education, administrative support that applicants must meet to receive accreditation. AAHRPP has accredited hundreds of organizations since 2001 (Association for Accreditation of Human Research Protections Programs 2012a, b). Accreditation can promote compliance with ethics policies and guidelines and signal the institution's commitment to integrity in research.

10.6 Conflict of Interest

A conflict of interest (COI) in research occurs when an investigator has a financial, professional or other interest which is likely to affect his or her tendency to fulfill his or her professional, ethical or legal obligations (Thompson 1993; Shamoo and Resnik 2015). COIs can impact adherence to professional, ethical, or legal duties by influencing judgement, decision-making, or behavior (Davis 1982; Resnik 2007a). COIs therefore increase the risk of misconduct, noncompliance, corruption, or other problems in research (Shamoo and Resnik 2015). Some examples of COIs include:

- **Financial**: ownership of stock of equity; intellectual property; receipt of speaking fees or honoraria from a research sponsor; paid consulting arrangements;
- **Professional**: relationships with students, mentors; collaborators;
- **Personal**: relationships with family of friends;
- **Political**: involvement in political organizations or interest groups.

Institutions may also have COIs if the organization or its leaders have financial or other interests which are likely to impact institutional decision-making. For example, if a private company is giving a major gift to a university or supporting millions of dollars of research, institutional leaders may make decisions that favor the company and compromise the university's academic mission or scientific integrity. Commentators on the Gelsinger case (discussed in Chap. 2) were concerned that the University of Pennsylvania had financial interests related to the research which could have impacted its oversight of the gene therapy experiment (Resnik 2007a).

A COI is different from a conflict of obligation or duty (Resnik 2007a). For example, a physician who is enrolling patients in a placebo-controlled clinical trial in which an effective therapy already exists has a conflict between his duties as a physician and his duties as a researcher, but not a COI, if he has no significant interests at stake. A conflict of obligation could become a COI if the investigator's interests favor one obligation over another.

A COI is also different from a conflict of commitment, which occurs when investigators have different professional commitments competing for their time and effort. For example, a university professor who spends a significant amount of time consulting for industry might not have enough time to meet her teaching, advising, or research obligations. Conflicts of commitment can usually be handled by proper time management. Most universities have policies concerning the amount of time and effort that one may devote to outside activities, such as private consulting (Shamoo and Resnik 2015).

COIs, especially financial ones, have increased in research since the 1980s, due to the growing commercialization of science and the complex web of interactions between academic institutions and private industry (Krimsky 2003). Two key pieces of legislation passed in the early 1980s, the Bayh-Dole Act and the Technology Transfer Act, allowed scientists to patent inventions developed using federal funds and to transfer those inventions to universities. Congress passed these laws to encourage transfer of technology from the public to the private sector and to spur economic development.

In the 1980s, private companies, especially pharmaceutical, biotechnology, and information technology firms, began investing more money in research and development (R &D) in their own laboratories and on academic campuses. At the same time, government support of non-military scientific research stagnated, due to budgetary constraints related to economic downturns and increased spending on defense and entitlement programs. R & D funding in the U.S. shifted from about 50% industry and 50% non-industry (government, university, and private philanthropy) in the 1970s to about 65% industry by the beginning of the twenty-first century (Shamoo and Resnik 2015).

In response to dwindling government support, universities have been seeking R & D contracts with private companies and accepting large corporate gifts, which they have used to develop research infrastructure. Today, many universities have buildings, laboratories, or endowed professorships named after industry donors and some have corporate campuses (Krimsky 2003). Universities also began to seek

income from intellectual property developed by faculty. Universities established technology transfer offices to help scientists apply for patents and to manage the institution's intellectual property, including copyrights. A university scientist who patents an invention is normally required to transfer the patent to the institution in exchange for a share of the royalties (usually around 50%). Universities also provided faculty with legal and financial support needed to form start-up companies to commercialize products and services (Shamoo and Resnik 2015). Faculty members began to rely more on private funding to support their research and they accepted money from industry for consulting, writing, or lecturing. Faculty members and universities also acquired stock or equity in private companies that sponsor R &D. Some universities formed private foundations to manage these investments and dole out venture capital for faculty companies (Krimsky 2003).

COIs are an ethical concern in human subjects research for several reasons. First, COIs may bias data or results (United States Congress 1990; Resnik 2007a; Shamoo and Resnik 2015). There is a growing body of evidence on the relationship between financial interests, industry funding, and research outcomes (Hampson et al. 2008). Numerous studies have demonstrated the existence of a funding effect in science: studies sponsored by the manufacturer of a product are more likely to favor the product than studies sponsored by an independent organization, such as a government agency (Bekelman et al. 2003; Sismondo 2008a). For example, Cho and Bero (1996) found that 98% of studies published in symposium proceedings with drug company support reported results favorable to the drug under investigation, compared to 79% without company support. Stelfox et al. (1998) found that among studies of calcium channel-blocking drugs with authors with ties to drug companies, 96% reported findings supportive of the drugs under investigation, while studies with authors without industry ties were neutral (60%) or critical (37%). Friedberg et al. (1999) found that 95% of articles with industry funding reported positive results for cancer treatments as opposed to 62% without industry funding. Ridker and Torres (2006) found that 65% of articles with industry funding reported favorable results for new cardiovascular drugs as opposed as opposed to 39.5% without industry funding. Studies have also shown that financial COIs are associated with research outcomes. Friedman and Richter (2004) reviewed 398 original articles published in the *New England Journal of Medicine* and the *Journal of the American Medical Association* in 2001 and found a strong association between one of more authors having a financial COI (such as stock or fees for consulting or speaking) and positive findings, e.g. evidence that a treatment is safe and effective. Articles that disclosed a financial COI were 2.3 times more likely to report positive findings than those that did not disclose a financial COI (Friedman and Richter 2004). Friedman and Friedman (2016) also found a strong association between financial COIs and research outcomes in 373 original articles published in occupational and environmental health journals in 2012. Articles disclosing financial COIs were 4.31 times more likely to report negative findings, e.g. no evidence that a chemical or agent is dangerous.

There are a several plausible explanations of the effect of financial interests on scientific research, i.e. the funding effect (Krimsky 2003). The first is that company

funding decisions can impact outcomes. For example, drug companies are required to conduct preliminary studies pertaining to a drug's chemistry, pharmacology, and toxicity before proceeding to clinical trials. Thus, when a company decides to fund a clinical trial it already has some data suggesting the drug may be safe and effective. Also, companies may decide to withdraw funding from studies that are not yielding favorable results. While it is important to counteract this potential source of bias, for example, by using public funds to sponsor independent studies, the fact that funding decisions made by companies can affect outcomes does not support the thesis that these outcomes are inherently tainted or unreliable (Resnik 2007a; Resnik and Elliott 2013).

A second explanation is that a company may decide not to publish results which go against its financial interests. Although many would regard data suppression as unethical under most circumstances, companies are not legally required to publish their research and can treat data and results as trade secrets. For example, if a company sponsors a clinical trial comparing its drug to competing drugs and the trial does not show that its drug is better than the competitors, it could decide to suppress the results of this study. It would still need to report the data to the FDA, but the FDA would treat the data as confidential. A salient example of this phenomenon occurred in 1995, when Boots Pharmaceuticals forced University of San Francisco pharmacologist Betty Dong to withdraw a paper from publication in the Journal of the American Medical Association. Boots had sponsored Dong's study, which compared its hypothyroidism drug Synthroid (levothyroxine) to generic alternatives. The study found that Synthroid was not superior to the generic drugs and that the U.S. government could save millions of dollars per year by switching Medicaid and Medicare patients with hypothyroidism from Synthroid to a cheaper generic drug. Dong had signed a contract with Boots giving the company the right to prevent her from publishing the research without the company's written approval. Dong withdrew her paper after the company threatened to sue her for violating the agreement. Fortunately, the company eventually decided to allow Dong to publish the study in the New England Journal of Medicine, but only after it had published an article favorable to Synthroid (Shamoo and Resnik 2015).

Sometimes a company's decision to suppress data can place human subjects at risk. For example, Nancy Olivieri and her collaborators at the University of Toronto (UT) and Toronto General Hospital (TGH) published a paper in the New England Journal of Medicine in 1995 which found that deferiprone is effective at treating the thalassemia and has few side effects. The drug's manufacturer, Apotex, sponsored their research. Olivieri soon noticed that many of her patients had increasing iron levels that placed them at risk of heart failure or death. In May 1996, she decided to inform TGH's research ethics board (REB) about this problem but the company told her not to do this. After discussing the issue with representatives from Apotex, Olivieri decided to submit a report to REB. Within three days, the Apotex terminated the research and withdrew all supplies of the drug from the hospital's pharmacy. The company also threatened to sue Olivieri if she told her patients, regulatory agencies, or researchers about her concerns with the drug. After receiving additional threatening letters from the company, Olivieri withdrew a presentation on the drug

she had submitted to a scientific meeting. In 1998, Apotex, which was in the middle of negotiation large donations to UT and TGH, pressured these institutions to take action against Olivieri, which they did. The institutions tried to discredit Olivieri and her research and took steps to dismiss her from her position. Fortunately, some of Olivieri's colleagues intervened on her behalf prevented her from being dismissed (Olivieri 2003).

A third explanation is that researchers or sponsors may sometimes deliberately bias results to promote their financial interests. Although published studies indicate that fraud is rare in science (Fanelli 2009), when it does occur significant financial interests are often at stake (Shamoo and Resnik 2015). For example, Poehlman admitted that financial pressures influenced his behavior. Wang was probably also impacted by the money and the prestige he expected to gain from his research. Most of the evidence concerning research misconduct in the U.S. has come from surveys of government-funded researchers or official findings of misconduct published by the NIH or NSF, and we know very little about misconduct cases, patterns, or trends in privately funded research (Shamoo and Resnik 2015). However, it is likely that the incidence of misconduct in privately-funded is not very different from the incidence in publicly-funded research.

A fourth explanation is that companies may decide not to publish all the data from a study. A notable example of data omission occurred in research on Merck's drug Vioxx (rofecoxib). In 1999, the FDA approved Vioxx for the treatment of arthritis and pain. Merck claimed that one of the benefits of the drug is that it produces fewer adverse gastrointestinal (GI) effects than other non-steroidal, anti-inflammatory drugs commonly used to treat arthritis and pain. Vioxx generated an estimated $2.5 billion in annual sales. In 2001, researchers published the results of a Merck-sponsored clinical trial, named VIGOR, in the New England Journal of Medicine. VIGOR showed that Vioxx produces fewer GI side effects than other drugs although it increases the risk of cardiovascular disease. The investigators, most of whom had financial ties to Merck, explained the difference in cardiovascular disease risks as due to the protective effects of the other drugs. However, the VIGOR study did not include all of the cardiovascular risk data collected by the researchers. If it had, it would have shown that Voixx patients had five times the risk of having a heart attack or stroke as compared to patients taking other medications. In 2002, the FDA notified Merck that it had misrepresented Vioxx's safety profile and required the company to include a black-box warning on the drug label. In 2002, another study of Vioxx was stopped early to protect patients receiving the drug from cardiovascular risks. In 2004, Merck withdrew Vioxx from the market due to safety and liability concerns. Thousands patients who had a heart attack or stroke while taking Vioxx have sued the company (Shamoo and Resnik 2015).

The same year that Merck withdrew Vioxx from the market, a systematic review found that several selective serotonin reuptake inhibitors (SSRIs), including Paxil (paroxetine), Zoloft (sertraline), and Prozac (fluoxetine) increase the risks of suicide when prescribed to children and adolescents. Previous published studies had not shown that these drugs increase the risk of suicide in children or adolescents. However, the systematic review included data that had been reported to the

Committee on Safety in Medicines (a U.K. agency similar to the FDA) but had not been previously published. Parents and psychiatrists were understandably furious when they found out that companies had suppressed this data, and New York Attorney General Eliot Spitzer sued the manufacturer of Paxil, Glaxo Smith Klein, for fraud (Shamoo and Resnik 2015).

A fifth explanation of the funding effect is that financial interests may exert subconscious influences on researchers' judgment which bias the results (Resnik and Elliott 2013). Scientific research involves many different decisions that require the exercise of judgment, such as choosing the study design, sample size, interventions, procedures, measurements, and statistical methods. For many of these choices, there may be more than one option that researchers would regard as legitimate. Scientists who have financial interests related to a study research may be unaware that they are making choices that tend to bias the study in a particular direction (Resnik 2007a). Studies have shown that financial interests can exert powerful, subconscious influences over human behavior. Even small gifts, such as pens and free lunches, can influence our decision-making at subconscious level (Katz et al. 2003).

COIs raise ethical concerns not only because they can increase the risk of bias but also because they may influence interactions between investigators and subjects during recruitment and consent. An investigator with a COI may be strongly motivated to recruit subjects for a study so he or she can generate results. The investigator may overemphasize the benefits of the study and downplay its risks to encourage an individual to participate. Commentators on the Gelsinger case were concerned that that PI of the study, James Wilson, may have failed to fully inform Gelsinger of the risks of the research to encourage him to enroll (Resnik 2007a). Even government funded investigators may have strong incentives to recruit participants to generate results and maintain research support (Resnik 2007a).

A third reason why COIs raise ethical concerns is that they may cause harm to human subjects or increase the risks of harm. For example, if Gelsinger had received more information about the risks of Wilson's gene therapy study, he might not have enrolled and could be alive today. In addition to the Gelsinger tragedy, another case demonstrated how financial interests may increase risks to human subjects. In the Olivieri case (discussed above) patients could have suffered serious harm if she had succumbed to pressure from Apotex to suppress safety data. Investigators with less courage and integrity than Olivieri may allow pressures from corporate sponsors or their own financial interests to influence decisions related to the safety of patients in clinical trials (Krimsky 2003).

A fourth reason why COIs raise ethical concerns is that they may compromise institutional oversight of research. For example, an IRB member with a conflict of interest related to a study may be predisposed to approve the study and not criticize it. Although federal research regulations require that IRB members not have any COIs related to the research they are reviewing, they do not define conflicts of interest (45 CFR 46.107e). Additionally, university leaders may place pressure on the IRB to approve a study that is expected to generate significant income for the institution via contracts or grants. Since IRB members often have relationships with

other investigators or administrators at an institution, it can be difficult avoid or manage COIs in the review of research with human subjects (Shamoo 1999c).

Finally, COIs raise ethical concerns because they may undermine trust in research, especially when they are not fully disclosed (DeAngelis 2000; Institute of Medicine 2009). Human subjects, communities, other investigators, and members of public may be concerned the COIs may influence judgment, decision-making, and behavior and lead to violations of professional, ethical, or legal standards. As noted in Chap. 2, the failure to adequately disclose the investigator's and the institution's financial interests in the Gelsinger incident was a major ethical concern in this tragedy, which adversely impacted trust in gene therapy research and the University of Pennsylvania. When research subjects or members of the public discover that a COI has not been fully disclosed, they may believe that investigators or institutions are trying to deceive them (Shamoo and Resnik 2015). For example, when Moore discovered that his physician had not told him that he was commercializing a cell line developed from his leftover tissue, he felt manipulated and deceived (see discussion in Chap. 2). COIs may undermine trust even when there is no evidence that they have impacted judgment, decision-making, or behavior because members of the public and other outside observers may perceive that they have biased research (Resnik 2007a).

Before Jesse Gelsinger's death in 1999, several other incidents highlighted how COIs can undermine the integrity of research and the public's trust. In the late 1980s, a U.S. congressional committee on government operations began investigating cases of research misconduct and conflict of interest in biomedical science (United States Congress 1990). One of these involved research on Retin A, a cream containing a vitamin A derivative (trans-retinoic acid) manufactured by Johnson and Johnson. The FDA had approved Retin A in 1971 for treating acne. In 1988, the *Journal of the American Medical Association* published a study claiming that Retin A is effective at treating photo-aged (e.g. wrinkled) skin (Weiss et al. 1988). An accompanying editorial touted the benefits of the medication for reversing the effects of aging on the skin (Gilchrest 1988). Shortly after the study was published, several television newscasts covered the story, and Johnson and Johnson's sales of the product skyrocketed (United States Congress 1990). Several of the authors of the study began plugging the compound at national conferences and in television and radio interviews. In 1989, an exposé published in *Money Magazine* revealed that several of the authors of the study and the author of the editorial had financial interests related to the research that had not been disclosed in print. One of the authors of the study, John Vorhees, had received several hundred thousand dollars in research support from Johnson and Johnson, and compensation for consulting and speaking (i.e. honoraria). The author of the editorial, Barbara Gilchrest, had received honoraria from Johnson and Johnson and her department had received several hundred dollars in research support. The magazine article also found that the authors had used lighting techniques to make the cream appear to be more effective than it actually was and that they had not used double-blinding, as they had claimed in the article (United States Congress 1990).

Two other noteworthy cases involved investigators profiting from sales of stock related to their research. In 1985, Scheffer Tseng published an article showing that an ointment containing Vitamin A is effective at treating dry eyes but he did not disclose that the FDA had granted him exclusive rights to market the product or that he held 530,000 shares in Spectra, a drug company which was planning to purchase those rights for $310,000. The price of Spectra's stock soared after the paper appeared in print and Tseng sold his shares. Subsequent research showed the ointment was not as safe or effective as Tseng had claimed (Resnik 2007a). In 1997, Michael Macknin published a study in the *Annals of Internal* Medicine showing that zinc lozenges are effective at treating the common cold. Macknin had disclosed to the editors of the journal that he owned 9000 shares in Quigley Corporation, the manufacturer of the lozenges, but the journal did not share this information with the readers. Shortly after the paper appeared in print, Macknin made a $145,000 profit from selling his stock. Subsequent studies published by Macknin and other researchers have shown that the zinc lozenges are no more effective than placebos at treating the common cold (Resnik 2007a).

An incident involving the NIH's intramural program raised concerns about the integrity of government scientists. In 2003, a series of articles published in the *Los Angeles Times* reported numerous violations of the NIH's rules concerning outside activities for intramural researchers and some lucrative consulting arrangements with private companies that apparently did not violate existing rules. Two Congressional committees began investigating these allegations and NIH Director Elias Zerhouni appointed a panel to examine the matter and recommend changes in the organization's ethics rules (Kaiser 2004). To restore Congress' and the public's trust in the integrity of NIH research, the organization adopted some of the world's strictest COI policies. The rules prohibit all NIH researchers from engaging in paid consulting for pharmaceutical and biotechnology companies and other substantially affected organizations (SAOs). The rules also prohibit NIH leaders from owning any stock in SAOs and place limitations on the amount of stock that other employees may own in SOAs (Resnik 2005b).

10.7 Data Suppression

As mentioned in Chap. 6, the sharing of data, methods, materials, and results is essential to the progress of science. In Chap. 6, we considered ethical dilemmas involving conflicts between openness and transparency and protection of privacy and confidentiality in research. As noted above, private companies often have strong financial reasons not to share or publish data. Companies are not legally required to publish their studies and usually treat them as trade secrets. Private companies may suppress data not only promote their products but also to their intellectual property rights. To obtain a patent on an invention, the new product or process must not have been previously disclosed in the scientific or technical literature or prior art (Miller and Davis 2011). Since publication of research related to a new invention can

invalidate the patent, a scientist or research sponsor may wait to publish or share data related to a new invention until they have obtained a patent. Also, if a scientist shares data or ideas related to a patentable invention with other researchers, his or her invention could be stolen (Resnik 2003a, b, c).

The issue of whether data suppression constitutes fraud or research misconduct is complex. Data suppression is ethically problematic because it is highly deceptive and could be construed as a data falsification, which the U.S. government defines as "manipulating research materials, equipment, or processes, or changing or omitting data or results such that the research is not accurately represented in the research record (Office of Science and Technology Policy 2000: 76262." In some cases, it may be appropriate to exclude data outliers from the analysis if they result from experimental or human error or are more than two standard deviations from the mean (Shamoo and Resnik 2015). To avoid deceiving scientists or the public about study outcomes, investigators should discuss data exclusions in their published research. To adhere to principles of honesty and transparency in research, an investigator who excludes data should explain why certain data points were not included and analyze the data with and without the outliers to show that removing these data points clarifies a statistical relationship among variables but does not impact the overall results (Shamoo and Resnik 2015). However, excluding data without providing a sound scientific justification, as may have occurred in the VIGOR study, may constitute data falsification (Shamoo and Resnik 2015). Not publishing all of the data from different studies is different from excluding data from a particular study because the data are from different datasets and do not necessarily address the same hypothesis or research question. Moreover, there may be some good scientific reasons, such as poor experimental design or inadequate statistical power, for not publishing data from a particular study. Selective publication of studies may constitute a pattern of deceptive even if it is not data falsification (Michaels 2008).

In the early 2000s, many biomedical journals began requiring registration as a condition of publishing the results of a clinical trial in an effort to minimize data suppression (Dickersin and Rennie 2003, 2012). Scientific journals that follow guidelines from the International Committee of Medical Journal Editors (ICMJE) also require clinical trial registration (Laine et al. 2007). In 2007, the U.S. Congress passed a law requiring sponsors or designated investigators to register clinical trials of FDA-regulated drugs, biologics, and medical devices, other than Phase I studies (ClinicalTrials.gov 2017). Clinical trial registration involves posting some basic information about a study on a public website, including:

- The title
- Goals, objective, and hypotheses
- Design, methods, procedures, interventions, drugs
- Population, inclusion and exclusion criteria
- Sponsor(s), investigator(s), institution(s)
- Contact information
- Outcomes and related publications

The world's largest clinical trial registration website is ClinicalTrials.gov, which contains records from over 200,000 studies conducted in 193 countries (ClinicalTrials.gov 2017).

Clinical trial registration can help counteract data suppression, but it cannot prevent it. Although sponsors are required to publish trial outcomes and related publications, they are not required to publish data. Thus, a sponsor could publish the results of a study but not all of the data. However, since there would be public record of the study, interested scientists could ask the investigator or sponsor for access to the data.

Another way of minimizing data suppression is for universities to negotiate contacts with corporate sponsors that do not allow them to unreasonably delay or block publication of research. Boots was able to block the publication of Dong's study because it had signed a contract with the University of San Francisco giving it the right to review and approve publications (Resnik 2007a).

10.8 Dealing with Conflicts of Interest in Research

There are several strategies for dealing with conflicts of interest in research. The most basic one is to disclose COIs to interested parties, such as institutions, IRBs, research participants, journals, granting agencies, etc. (Institute of Medicine 2009; Shamoo and Resnik 2015). Disclosure embodies the virtues of honesty, openness, and transparency and allows interested parties to decide how to respond to the COI. Readers of journals articles may use COI disclosures to assess the credibility of the research (Resnik and Elliott 2013). A scientist who learns that the authors of a paper have significant financial relationships to a company that sponsored their research may decide to give the paper additional scrutiny to ascertain the validity of the data, methods, and approach. Human research subjects may consider COI disclosure relevant to their decision to enroll in a study (Resnik 2004). Jesse Gelsinger might not have participated in the study that led to his death if he had known the extent of the investigator's and university's financial COIs. Studies have found that most research participants or potential participants would like to receive information concerning financial interests and sources of funding related to the research (Kim et al. 2004; Grady et al. 2006; Coors et al. 2015).

Disclosure can also promote trust by assuring interested parties that significant interests are out in the open. Failing to disclose COIs can destroy trust, since parties that discover undisclosed interests may feel that the person with the COI has been trying to deceive them (Shamoo and Resnik 2015). While some have argued that disclosing COIs to human subjects may make them suspicious of investigators' motivations and undermine their trust in research (Thompson 1993), studies indicate that disclosure can have the opposite effect if done properly (Levinson et al. 2005; Coors et al. 2015). COIs disclosures should be expressed in non-technical language understandable to the subjects (Resnik 2004).

Granting agencies, such as the NIH and NSF, require researchers to disclose significant financial interests to the agency and their institution. The NIH, for example, requires funded extramural investigators or those seeking funding to disclose financial interests greater than $5000 to the agency and institution (Department of Health and Human Services 2011). The NIH and NSF both require institutions to establish committees to receive and manage COI disclosures (Shamoo and Resnik 2015). As noted above, the NIH also requires intramural researchers to disclose COIs and has oversight mechanisms for managing them. Most scientific journals also require authors to disclose financial or other interests related to submitted manuscripts (Cooper et al. 2006; Ancker and Flanagin 2007). IRBs also require investigators to disclose COIs to the committee and to human subjects (Shamoo 1999a; Shamoo and Resnik 2015).

While disclosure is a key strategy for dealing for dealing with COIs, it may not be sufficient to promote integrity and trustworthiness in many cases (Elliott 2008). Strategies that go beyond mere disclosure include avoidance and management (Shamoo and Resnik 2015). Avoidance involves taking steps to prevent the conflict from occurring, such as prohibiting or restructuring the situation that gives rise to the conflict. For example, the NIH and NSF prohibit individuals from reviewing research proposals from current or former (within the last 3 years) students, collaborators, or advisors (Shamoo and Resnik 2015). As mentioned above, the federal regulations forbid IRB members reviewing studies in which they have COIs. IRBs may allow a member with a COI to share information about the study with the committee but prohibit him or her from taking part in discussion or voting (Wolf and Zandecki 2007). Journal editors may ask peer reviewers not to evaluate manuscripts in which they have COIs (Shamoo and Resnik 2015).

Managing the conflict of interest may be the best option when avoidance is not feasible or it likely to have adverse consequences for science or society. For example, suppose that an investigator has patented a tool to be used in robotic surgery and she is planning to conduct a clinical trial of the device at an academic medical center. The investigator has also formed a startup company to develop, test, manufacture, and market the device. She also owns stock in the company. The investigator transferred the patent to the institution but will receive a share of royalties from it. The startup company will license the device from the institution. The institution also owns stock in the company. In this situation, there are numerous individual and institutional COIs that could affect the integrity and trustworthiness of the research. The COI committee reviewing this situation could mandate that the investigator and institution avoid the conflict by requiring the investigator to conduct the clinical trial at another institution or by requiring the investigator and institution to sell their stock in the company. However, this proposal could undermine product development and research. The investigator might be the best person to conduct and manage the study, and moving the study to another institution might compromise the quality of the research. If the investigator and institution sell their stock in the company, the company may suffer significant losses and go bankrupt. The best way to handle this situation might be to allow the study to take place on campus but require additional oversight of the research to ensure that human subjects are protected, that the

research is well-designed and properly implemented, and that data and results are published in a timely fashion (Shamoo and Resnik 2015).

Institutional COIs can be difficult to manage because there may not be an individual or committee within the organization with the authority or expertise to make fair and unbiased decisions concerning the institution's financial other interests (Resnik and Shamoo 2002). The Board of Trustees or similar body is responsible for ensuring that the institution achieves its academic, social, and economic goals and complies with ethical and professional standards. The Board could establish a committee to review and oversee COIs pertaining to the institution and its leaders. Other steps that institutions can take to manage their own COIs include separating research oversight from financial decision-making. For example, IRBs and compliance officers should be able to make decisions which are not influenced by the interests of the institution or its leaders. Institutions can also create foundations to hold institutional investments and intellectual property rights (Moses and Martin 2001).

To decide whether avoidance or management is the best way to handle a COI, one should consider the following factors (Institute of Medicine 2009; Shamoo and Resnik 2015):

- The strength of the conflict: how likely is it that the conflict will lead to bias, noncompliance, or other ethical or legal problems in research? How much money is at stake? What types of relationships, financial or otherwise, are involved?
- The difficulty of managing the conflict: does the institution have the resources to manage the conflict? Can the institution create special committees or other mechanisms to manage the conflict?
- The consequences of avoiding the conflict: what are the likely consequences of avoiding the conflict for human subjects, the investigator, institution, science, and society? Will avoiding the conflict prevent important research or product development from taking place? Will not avoiding the conflict undermine public trust?

Some scholars have proposed more radical ways addressing relationships between private companies and academic institutions. James Robert Brown (2000, 2002) has argued that the best way to minimize bias, corruption, secrecy, and other ethical problems related to financial interests in research is to sever the relationship between academic institutions and private corporations: Academic research should be funded by government grants, not private money. Brown also argues that biomedical patenting should cease so that investigators will share data more freely and drugs, biologics, other medical products and technologies will be accessible to public at low cost. Brown's solution to the problem of financial interests is to socialize academic research.

Brown's proposal is unrealistic and counterproductive. First, as noted earlier, private industry supports about 65% of R & D and governments are not likely to have the funds pay for research at current levels if one follows Brown's proposal. Brown's proposal would radically reduce the amount of research conducted on academic campuses and increased the amount conducted in private laboratories or corporate facilities. Since companies protect research under trade secrecy laws, shifting

research from academic campuses to private venues is not likely to increase openness, data sharing, or publication in science. Indeed, it would probably have the opposite effect. Moreover, purging private research funding from academic institutions is not likely to eliminate conflicts of interest, bias, or other problems, since ethically vexing situations can still arise when scientists compete for government funds (Resnik 2007a).

Second, foregoing biomedical patenting would have many adverse consequences for the development of drugs, biologics, medical devices, and other biotechnological inventions. Since the 1700s, the overarching rationale for the patent law has been to stimulate science and the practical arts (Resnik 2003b; Miller and Davis 2011). Prior to the advent of the patent system, scientists and companies relied on secrecy to protect their intellectual property. Governments started granting patents to encourage inventors to share their trade secrets with the public. In exchange for this disclosure, government allowed inventors to monopolize their work for a limited time. Under U.S. law, a patent grants the patent owner the right to exclude others from using, making, or commercializing his or her invention for 20 years from the date of the application (Miller and Davis 2011). The patent holder must submit an application to the patent office describing the invention and how to make and use it. Once the patent is awarded, the application becomes public (Miller and Davis 2011). Patenting provides economic incentives not only for inventors but also for investors. Private companies invest in R & D in order to profit from patented products or methods. Indeed, pharmaceutical companies derive most of the profits from new drugs when they are still under patent. While patented drugs are more expensive than drugs not protected by patents, once a patent expires prices drop precipitously, since other companies can manufacture and sell the drug, which introduces competition to the market (Resnik 2003a, b, c). Although patenting encourages secrecy while scientists are working on patentable research, publication and data sharing can take place when the patent is awarded. Overall, the benefits of patenting for biomedical science, biomedicine, and society outweigh the harms (Resnik 2003a, b, c).

Third, academic researchers and institutions benefit from collaborations with industry. In addition to receiving money to support research, academic scientists and institutions also benefit from exchanging information, knowledge, and ideas with their colleagues in industry (Resnik 2007a; Shamoo and Resnik 2015). They have the opportunity to work with companies who are developing cutting-edge products, methods, and technologies. Students can benefit from learning about real-world problems, internships with private companies, and employment opportunities in the private sector after graduation. Society can also benefit from collaborations between academic and industry that lead to the development of useful products, services, and technologies.

Others have made more modest proposals for dealing with relationships between academic institutions and private companies (Moses and Martin 2001; Morin et al. 2002; Krimsky 2003; Resnik 2007a; DeAngelis and Fontanarosa 2008; Angell 2008). Some of these, such as clinical trial registration, conflict of interest management, and review of contracts with companies to allow for publication, were discussed earlier. Others include:

- Increasing government sponsorship of clinical research to provide a source of data independent from industry. For example, the government could sponsor studies designed to compare different treatments available on the market so physicians and patients will have information concerning the comparative benefits, risks, and costs of these treatments (Resnik 2007a).
- Strengthening authorship criteria used by journals to clearly prohibit ghost authorship of scientific papers (DeAngelis and Fontanarosa 2008, see discussion Chap. 2).
- Empowering journals to deal with undisclosed COIs. A journal which discovers that an author has failed to disclose a significant financial or other interest related to his or her research could prohibit the author from publishing in the journal for a period of time (Goozner 2004).
- Coordinating COI management at institutions. For example, faculty COI committees should provide IRBs with disclosure information. Likewise, IRBs could share information relevant to COI assessment with faculty committees.

10.9 Conclusion

In this chapter I have examined several research integrity issues which have a direct bearing on protection of human subjects. These problems are important to address because they can increase risks to subjects, reduce benefits, undermine informed consent, compromise data, and damage the trustworthiness of research. Because investigators are not perfect, deliberate misbehavior, subconscious bias, and honest mistakes are likely to occur in research involving human subjects. Academic institutions, public and private research sponsors, and scientific journals should develop policies, procedures, administrative structures, and educational programs to prevent these problems and deal with them effectively and fairly when they occur. Leadership, mentoring, auditing, and accreditation can also play a key role in promoting research integrity. In the next chapter, I will move beyond issues impacting individuals and institutions and explore ethical and policy questions related to oversight of research with human subjects at the national level.